

Module 4: Biodiversity, evolution and disease

In this module the learners study the biodiversity of organisms; how they are classified and the ways in which biodiversity can be measured. It serves as an introduction to ecology, emphasising practical techniques and an appreciation of the need to maintain biodiversity. The learners also gain an understanding of the variety of organisms that are pathogenic and the way in which plants and animals have evolved defences to deal with disease. The

impact of the evolution of pathogens on the treatment of disease is also considered.

The relationships between organisms are studied, considering variation, evolution and phylogeny.

Learners are expected to apply knowledge, understanding and other skills developed in this module to new situations and/or to solve related problems.

4.1 Communicable diseases, disease prevention and the immune system

4.1.1 Communicable diseases, disease prevention and the immune system

Organisms are surrounded by pathogens and have evolved defences against them. Medical intervention can be used to support these natural defences.

The mammalian immune system is introduced.

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the different types of pathogen that can cause communicable diseases in plants and animals	To include, • bacteria – tuberculosis (TB), bacterial meningitis, ring rot (potatoes, tomatoes) • viruses – HIV/AIDS (human), influenza (animals), Tobacco Mosaic Virus (plants) • protist – malaria, potato/tomato late blight • fungi – black sigatoka (bananas), ringworm (cattle), athlete's foot (humans).

(b) the means of transmission of animal and plant communicable pathogens	To include direct and indirect transmission, reference to vectors, spores and living conditions – e.g. climate, social factors (no detail of the symptoms of specific diseases is required).
	<i>M0.1, M0.2, M0.3, M1.1, M1.2, M1.3, M1.5, M1.7, M3.1, M3.2</i> HSW1, HSW2, HSW3, HSW5, HSW6, HSW7, HSW8, HSW11, HSW12
(c) plant defences against pathogens	To include production of chemicals AND plant responses that limit the spread of the pathogen (e.g. callose deposition).
(d) the primary non-specific defences against pathogens in animals	Non-specific defences to include skin, blood clotting, wound repair, inflammation, expulsive reflexes and mucous membranes (no detail of skin structure or all the steps involved in the clotting cascade are required).
	HSW2, HSW8
(e) (i) the structure and mode of action of phagocytes (ii) examination and drawing of cells observed in blood smears	To include neutrophils and antigen-presenting cells AND the roles of cytokines, opsonins, phagosomes and lysosomes.
	PAG1 HSW4, HSW8
(f) the structure, different roles and modes of action of B and T lymphocytes in the specific immune response	To include the significance of cell signalling (reference to interleukins), clonal selection and clonal expansion, plasma cells, T helper cells, T killer cells and T regulator cells.
	HSW8
(g) the primary and secondary immune responses	To include T memory cells and B memory cells.
	<i>M1.3</i> HSW2
(h) the structure and general functions of antibodies	To include the general protein structure of an antibody molecule.
(i) an outline of the action of opsonins, agglutinins and anti-toxins	
(j) the differences between active and passive immunity, and between natural and artificial immunity	To include examples of each type of immunity.
(k) autoimmune diseases	To include an appreciation of the term <i>autoimmune disease</i> and a named example e.g. arthritis, lupus.

(l) the principles of vaccination and the role of vaccination programmes in the prevention of epidemics

To include routine vaccinations

AND

reasons for changes to vaccines and vaccination programmes (including global issues).

M0.1, M0.2, M0.3, M1.1, M1.2, M1.3, M1.5, M1.7, M3.1, M3.2

HSW1, HSW2, HSW3, HSW5, HSW6, HSW7, HSW8, HSW9, HSW11, HSW12

(m) possible sources of medicines

To include examples of microorganisms and plants (and so the need to maintain biodiversity)

AND

the potential for personalised medicines and synthetic biology.

HSW7, HSW9, HSW11, HSW12

(n) the benefits and risks of using antibiotics to manage bacterial infection.

To include the wide use of antibiotics following the discovery of penicillin in the mid-20th century

AND

the increase in bacterial resistance to antibiotics (examples to include *Clostridium difficile* and MRSA) and its implications.

HSW2, HSW5, HSW9, HSW12